INCH-POUND

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SUPERSEDING
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PERFORMANCE SPECIFICATION SHEET

ELECTRON TUBE, POWER
TYPE 7C23

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the electron tube described herein shall consist of this document and the latest issue of MIL-PRF-1.

DESCRIPTION: Triode

Outline: See figure 1.
Mounting position: Vertical, anode down.
Weight: 5 1/4 pounds nominal.

ABSOLUTE-MAXIMUM RATINGS:

| Parameter: | F1 | Ef | eb | Ec | Ib | Pp |
|---|---|---|---|---|---|---|
| Unit: | MHz | V ac | kv | V dc | A dc | W |
| Maximum: | 50 | 11.55 | 15 | -1500 | 1.25 | 1200 |
| Minimum: | - - - | 10.45 | - - - | - - - | - - - | - - - |
| Test conditions: | - - - | 11.0 | 3.5 | - - - | - - - | - - - |

GENERAL:

Qualification - Required.


DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

TABLE I. Testing and Inspection.

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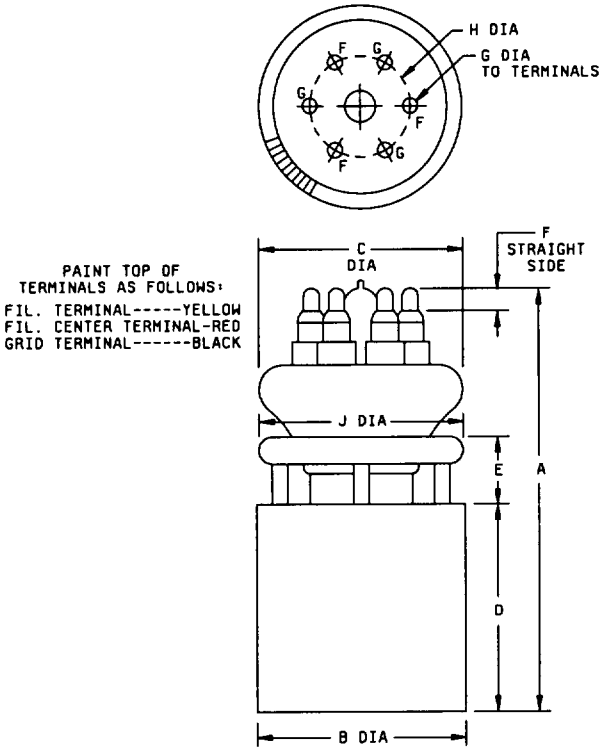
| Inspection | Method | Conditions | Acceptance levels | Inspection level or code | Symbol | Limits Min | Limits Max | Unit |
|---|---|---|---|---|---|---|---|---|
| Conformance inspection, part 1 | | | | | | | | |
| Filament current | 1301 | | .65 | II | If | 27.5 | 30.5 | A ac |
| Electrode current (grid) | 1256 | Ec/Ib = 0.35 A dc; t = 180 (see note 3) | .65 | II | Ic | - - - | 15 | µA dc |
| Electrode voltage (anode) | 1261 | Ec = 0; Ib = 0.2 A dc | .65 | II | Eb | 1.6 | 2.1 | kV dc |
| Electrode voltage (anode) | 1261 | Ec = -50 V dc; Ib = 0.2 A dc | .65 | II | Eb | 2.6 | 3.2 | kV dc |
| Amplification factor | 1316 | Ec/Ib = 0.2 A dc | .65 | II | Mu | 20.0 | 26.5 | - - - |
| Electrode voltage (grid) | 1261 | Ec/Ib = 100 µA dc; Eb = 15kV dc | .65 | II | Ec | - - - | -1000 | V dc |
| Peak emission | 1231 | ec = eb = 4,000 V (see note 3) | .65 | II | is | 25 | - - - | a |
| Insulation | - - - | Rp = 300; Ib = 4 to 6 mA dc; Eb = 20 kV dc (see notes 2 and 3) | .65 | II | - - - | - - - | - - - | - - - |
| Power output | - - - | Ib = 0.75 A dc; Ic = 0.15 A dc; Rg = 2,000 F = 2 MHz | .65 | II | Po | 1500 | - - - | W (useful) |
| Conformance inspection, part 2 | | | | | | | | |
| Direct-Interelectrode capacitance | 1331 | | - - -<br>- - -<br>- - - | - - -<br>- - -<br>- - - | *Cgp*<br>Cgf<br>Cpf | 11.4<br>12.6<br>1.2 | 15.0<br>16.4<br>2.2 | pF<br>pF<br>pF |
| Conformance inspection, part 3 | | | | | | | | |
| Life-test provisions | - - - | Group D, Power output; t = 500 hours | - - - | - - - | - - - | - - - | - - - | - - - |
| Life-test end point | - - - | Peak emission | - - - | - - - | is | 20 | - - - | a |

NOTES:

1. Tube may be used under pulse conditions in which anode voltage may be exceeded for certain conditions.
2. Anode potential is slowly increased to 20 kV dc at which potential the time without flashovers shall be less than 300 seconds. A capacity of not less than 0.1 mfd shall be connected in parallel with the anode supply.
3. Test shall be performed at the conclusion of the holding period.



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| Dimensions | | | | |
|---|---|---|---|---|
| | Inch | | Millimeter | |
| Ltr | Min | Max | Min | Max |
| Conformance inspection, part 2 | | | | |
| A | 6.50 | 7.00 | (165.1) | (177.8) |
| B | 3.469 | 3.531 | (88.11) | (89.69) |
| D | 3.375 | 3.625 | (85.73) | (92.08) |
| Conformance inspection, part 3 (periodic-check) | | | | |
| C | | 3.25 | | (82.55) |
| E | 0.875 | 1.000 | (22.23) | (25.40) |
| F | 0.375 | | (9.53) | |
| G | 0.359 | 0.391 | (9.12) | (9.93) |
| H | 1.375 | 1.625 | (34.93) | (41.28) |
| J | | 3.75 | | (95.25) |

FIGURE 1. Outline drawing of electron tube type 7C23.

Custodians:
Navy - EC

Review activities:
Navy - AS, CG, MC, OS

Preparing activity:
DLA - CC

(Project 5960-N227)



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